Introduction to Importance Sampling

"Brute Force. vs. Gillespie" worksheet.

Metropolis Acceptance Criteria in Monte Carlo Simulations.

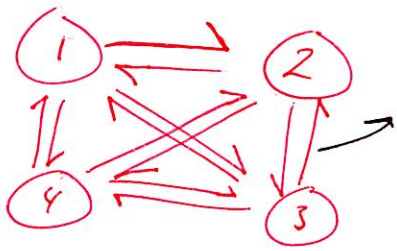
We want to generate states of a system that follows a Boltzmann Distribution: $\boxed{\pi(x) \propto e^{-\beta E(x)}}$

How do we do this? If we don't know the partition function, we don't know what the actual probability should be!

↳ We <u>compare</u> probabilities

A system at equilibrium can satisfy ~~detai~~ <u>detailed balance:</u>

Detailed balance states that the "flow" between any two states is equal and opposite.



$\text{rate}(2 \to 3) = \text{rate}(3 \to 2)$

Another way to think about this is that the probability you are in a state multiplied by the probility you transition to another state should be equal and opposite.

$$\pi(x) P(x \to x') = \pi(x') P(x' \to x)$$

The Metropolis Algorithm generates the ensemble of states in the probability distribution $\pi(x)$ by proposing moves and then accepting or rejecting moves based on the probability distribution.

That is, we propose going from x to x' some ~~function~~ probability $g(x \to x')$. We ~~accept~~ accept this move based on the ratio of the probability of x' relative to x

$$P_{Accept}(x \to x') = \frac{\pi(x')}{\pi(x)}.$$

[Why? What does this help us achieve?]

To satisfy detailed balance, ~~the~~ we must consider the probability of proposing moves:

$$P_{Accept}(x \to x') = \frac{\pi(x')\, g(x' \to x)}{\pi(x)\, g(x \to x')}.$$

But if $P_{Accept}(x \to x') > 0$ and < 1, then $P_{Acc}(x' \to x) > 1$, so to normalize the probabilities we use the form

$$\boxed{P_{Accept}(x \to x') = \mathrm{Min}\left[1, \frac{\pi(x')\, g(x' \to x)}{\pi(x)\, g(x \to x')}\right]}$$

Hastings did pioneering work on generalizing the idea of proposal moves but in general we can use a symmetric scheme $g(x' \to x) = g(x \to x')$ to simplify our fraction.

Can you think of schemes for this?